MODULE 03 · PROGRAMMING

Python Programming

A Custom AI-Generated Companion Guide

A clean, readable language used everywhere from automation to data science to AI.

DIGITAL GIRLS INITIATIVE PAKISTAN
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8–10 weeks · 5–7 hrs/week

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Welcome & How to Use This Guide

Python is often recommended as a first programming language because its syntax reads almost like English — and it's the same language powering data science, automation, and artificial intelligence at a professional level. This track takes you from your first print statement to building small real programs.

> **Mentor Tip**
> Python's official style guide is called PEP 8 — skim it once early on so your code looks professional from the start.

Estimated time: 8–10 weeks · 5–7 hrs/week · Level: No experience needed

01 Python Fundamentals

Syntax, variables, and control flow

Get comfortable with the building blocks every Python program shares, before moving on to anything more complex.

01 Set up Python
Install Python and choose an editor — VS Code or a browser-based notebook both work.

02 Variables & data types
Strings, integers, floats, and booleans.

03 Input & output
print() and input() — talking to the person running your program.

04 Conditionals & loops
if/elif/else, for loops, and while loops.

★ Practice Project
Build a simple calculator that takes two numbers and an operator from the user.

02 Data Structures & Functions

Organizing information and logic

This phase covers how Python stores collections of data, and how to package logic into reusable functions.

01 Lists & tuples
Ordered collections, indexing, and slicing.

02 Dictionaries & sets
Key-value pairs and unique collections.

03 Writing functions
def, parameters, return values, and default arguments.

04 Modules
Importing Python's built-in library instead of reinventing it.

★ Practice Project
Build a contact book stored as a list of dictionaries, with add / search / delete functions.

03 OOP & File Handling

Structuring bigger programs

Real programs are built from objects, and often need to read and write files — this phase introduces both.

01 Classes & objects
Modeling real-world things as code — the basis of most large Python projects.

02 File input/output
Reading from and writing to .txt and .csv files.

03 Error handling
try / except so your program fails gracefully instead of crashing.

04 Using external libraries
Installing packages with pip and reading their documentation.

★ Practice Project
Build a personal expense tracker that saves entries to a file and calculates totals by category.

04 Applied Python

Real, useful mini-projects

This phase is about applying everything you've learned to programs that solve an actual problem.

01 Working with APIs
Requesting live data from the internet using the requests library.

02 Simple automation
Writing a script that automates a repetitive task — renaming files, for example.

03 Intro to data with Python
A first look at the pandas library for working with tabular data.

04 Code review habits
Reading and cleaning up your own earlier code — an underrated skill.

★ Practice Project

Build a quiz app that pulls questions from a file or API, tracks score, and shows results at the end.

Free Video Courses

Curated, highly-rated, and completely free. Pick whichever teaching style fits you best.

> **CS50P — Harvard's Intro to Programming with Python**
> University-grade course, free on edX and YouTube
> *https://cs50.harvard.edu/python/*

> **freeCodeCamp — Python for Beginners**
> A complete, structured beginner course
> *https://www.freecodecamp.org/news/tag/python/*

> **Corey Schafer — Python Tutorials**
> Widely considered the clearest Python teacher on YouTube
> *https://www.youtube.com/@coreyms*

> **Kaggle Learn — Python Micro-Course**
> Short, hands-on lessons in your browser, no setup
> *https://www.kaggle.com/learn/python*

Your Practice Checklist

Print this page or copy it into a notebook. Check off each project as you finish it — real, working projects are what turn this guide into a portfolio.

- [ ] **Phase 01: Python Fundamentals — Build a simple calculator that takes two numbers and an operator fr**
- [ ] **Phase 02: Data Structures & Functions — Build a contact book stored as a list of dictionaries, with add**
- [ ] **Phase 03: OOP & File Handling — Build a personal expense tracker that saves entries to a file and calc**
- [ ] **Phase 04: Applied Python — Build a quiz app that pulls questions from a file or API, tracks score**

Finished all four phases?

Bring your projects to a DG·IP mentor session — we review portfolios and help you plan your next module.

You've got the roadmap.
Now go build something.

This guide is one part of the free DG·IP curriculum. Come back to the website for the live version of this module, curated videos, and seven other free skill tracks.

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